

CSE330 Spring 2026 Dashboard

[Instructor Details](#)

Saadat Rafid Ahmed [SADF/SRD]
Desk: 4J51
Email: saadat.ahmed@bracu.ac.bd

Schedule & Consultation Slot Booking

 Book Consultation Appointment [SADF]

Communication

All communication will be done through discord public channel and announcements will be made through notion.

You may do not DM via Discord for direct communication. Email instead.

Lecture Videos

- ▼

AQU Sir’s Materials

 [Select file to preview](#)

 [Google Drive Mid - Google Drive](#)
- ▼

RRH Sir’s Materials

 [CSE 330 - Supplementary Lectures-RRH](#)
- ▼

NLH Videos

 [Niloy Farhan CSE330 Summer 2025](#)
- ▼

BuX Videos

 BUX Videos - CSE330
Google Docs Embed

Course Outline and Materials

Textbook - [Main Anthony Yeats](#)

Course Overview - [overview link](#)

Mark Distribution

Assessment Tools	Weightage (%)
Attendance	5
Quiz	15 (n-2)
Assignment	10 (n-1)
Midterm	25
The mandatory final exam	25
Lab	20

Mark Distribution

Midterm Syllabus (According to the Lecture Note):
Chapter-1: 1.1, 1.2, 1.3, 1.4, 1.5
Chapter-2: 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7
Chapter-3: 3.1, 3.2, 3.3

Final Exam Syllabus (According to the Lecture Note):
Chapter-4: 4.1, 4.2, 4.4, 4.6, 4.7
Chapter-5: 5.1, 5.2, 5.3, 5.4
Chapter-6: 6.1, 6.2, 6.3
Chapter-7: 7.1, 7.2, 7.3

▼ Detailed Useful Tricks

- Check these videos for fraction binary to decimal conversion.
Manual Process: <https://www.youtube.com/shorts/Ph7x1MXGSsU>
Calculator: <https://www.youtube.com/shorts/cV9AiVZy bq0>

- You may refer to the following videos for finding inverse of a matrix, determinant of a matrix and for matrix multiplication using a calculator. These will be helpful for this course.
 - Inverse of a Matrix:
991-ES: <https://youtu.be/AFV6kUo6dls?feature=shared>
991-EX: <https://youtu.be/sbFT0aNsoiU?feature=shared>

 - Multiplying two matrices:
991-ES: <https://youtu.be/75NnTVDKHNA?feature=shared>
991-EX: <https://youtu.be/Tfj4RDfzjCA?feature=shared>

 - Determinant of a Matrix:
991-ES: <https://youtu.be/qK8HQiwoBPg?feature=shared>
991-EX: <https://youtu.be/4gVCk4Vbg6c?feature=shared>

 - How to find derivative in casio fx-991es plus:
<https://www.youtube.com/shorts/6QQf17Hu1Pw?feature=share>
Substitute X Values for equation: <https://youtu.be/GPVXrqehjRE>
Solving for X in an equation: <https://youtu.be/-Jkezu5nAlM>
Solving quadratic equation: <https://www.youtube.com/shorts/LMMMyVmFEQoU?feature=share>
Solving linear system: <https://youtu.be/rKlJmYdrGDo>

 - Use Radian mode for trigonometry, calculus, and arc length calculations in this course.

▼ Student Attendance Report - Theory Section 12 & 13

 Google Docs Saadat Rafid Ahmed - Spring 2026 - CSE330

Saadat Rafid Ahmed - Spring 2026 - CSE330

1321	24121378	Irfan Pradhan	CS	irfan.pradhan@g.bracu.ac.bd	1793887357	CSE330	13	OTHER	S
1322	24201462	Syeda Faiza Jarin	CSE	syeda.faiza.jarin@g.bracu.ac.bd	1609881190	CSE330	13	OTHER	S
1323	24241216	Khandkar Sajiduzzaman	CS	khandkar.sajiduzzaman@g.bracu.ac.bd	1633086782	CSE330	13	OTHER	S
1324	24301001	Md Raiyan Jalil Abir	CSE	raiyan.jalil.abir@g.bracu.ac.bd	1716322582	CSE330	13	OTHER	S
1325	24301038	Moshiour Rahman Mohib	CSE	moshiour.rahman.mohib@g.bracu.ac.bd	1905713688	CSE330	13	OTHER	S
1326	24301213	Mohd. Tirtho Rahman	CSE	mohd.tirtho.rahman@g.bracu.ac.bd	1300726817	CSE330	13	OTHER	S
1327	24301219	Md. Ishtiaq Mozumder	CSE	md.ishtiaq.mozumder@g.bracu.ac.bd	1511696062	CSE330	13	OTHER	S
1328	24301406	Swagata Chakraborty	CSE	swagata.chakraborty@g.bracu.ac.bd	1794314466	CSE330	13	OTHER	S
1329	24301415	Fahim Faisal	CSE	fahim.faisal5@g.bracu.ac.bd	1540051306	CSE330	13	OTHER	S
1330	24301433	Namira Hossain	CSE	namira.hossain@g.bracu.ac.bd	1311931263	CSE330	13	OTHER	S
1331	24301466	Fahim Nasir Mahmud	CSE	fahim.nasir.mahmud@g.bracu.ac.bd	1836999992	CSE330	13	OTHER	S
1332	24301536	Tasnim Shahadat Nuha	CSE	tasnim.shahadat.nuha@g.bracu.ac.bd	1775531163	CSE330	13	OTHER	S
1333	24301551	Muqtadir Hossain Khan	CSE	muqtadir.hossain.khan@g.bracu.ac.bd	1748297210	CSE330	13	OTHER	S
1334	24321265	Rakibul Hasan Shihab	CS	rakibul.hasan.shihab@g.bracu.ac.bd	1537147034	CSE330	13	OTHER	S
1335	24321284	Sahana Khan	CSE	sahana.khan@g.bracu.ac.bd	1734616854	CSE330	13	OTHER	S

Class, Assignment and Quiz Deadline Details

Assignments

- All assignments are handwritten
- No submission will be counted unless both the hard copy and soft copy is submitted
- If found cheating, all involved will be penalised
- Late submission will not be counted

Assignment Submission Form: <https://forms.gle/pMqRtSsNhgDBbHnq9>

Assignment 01

Question Link: <https://drive.google.com/file/d/1RFaeyGmV3Pt4qnSzLrso2QybOwKLlw1c/view?usp=sharing>

Deadline:
1 March 2026 11:59 PM (Online)
2 March 2026 (Hard Copy-During Class Only)

Assignment 02

Quesiton :

https://drive.google.com/file/d/17OLc_6Z-5pojMtCTfvIoe8POYRXWZYxy/view?usp=drive_link

Deadline: 27 April 2026 Both Hardcopy and SoftCopy

Assignment 03

Question: https://drive.google.com/file/d/1yZzVWVbdoM63CKoZY-Rxnj_Zyw9yk6S/view?usp=drive_link

Deadline: 4 May 2026 Both Hardcopy and SoftCopy

Assignment 04

Question: https://drive.google.com/file/d/1Uz372t521rKy9m7eXCt7ExYvj6rcoR7b/view?usp=drive_link

Deadline: 15 May 2026 - only online submission

Quizzes

Quiz 01

Date: 18 Feb 2026

Topics

- 1.1.1
- 1.1.2
- 1.1.3
- 1.1.4
- 1.2.1
- 1.2.2

▼ Class Details

 Google Docs Saadat Rafid Ahmed - Spring 2026 - CSE330

Saadat Rafid Ahmed - Spring 2026 - CSE330 : Class Schedule										
Number	Date MON/WED	Section12	Section13	Quiz	Topics	Assignments		Chapter	Topic Sr.	Pre Requisite
1	02Feb	☒	☒		1.1.1 1.1.2		PREMID	1	1.1.1	
2	9Feb 	☒	☒		1.1.3 1.1.4			1	1.1.2	
3	16Feb 	☒	☒		1.2.1 1.2.2			1	1.1.3	
4	18Feb	☒	☒	Q1 Topics 1.1.1 1.1.2 1.1.3 1.1.4 1.2.1 1.2.2	1.3.1 1.3.2	Lab 02  Assignment_01_Spring_2026.pdf		1	1.1.4	
5	23Feb	☐	☐		1.3.1 1.3.2 2.1 2.2			1	1.2.1	
6	25Feb	☐	☐					1	1.2.2	
7	02Mar	☐	☐	Q2				1	1.3.1	
8	04Mar	☐	☐					1	1.3.2	
9	09Mar	☐	☐							
10	11Mar	☐	☐					2	2.1	TRUE
11	16Mar	☐	☐	Q3				2	2.2	
								2	2.3	
								2	2.4	
								2	2.5	
								2	2.6	
								2	2.7	

Class Schedule

 Saadat Rafid Ahmed - Spring 2026 - CSE330
Google Docs Embed

Chapter Details

1 Chapter 1: Floating Point Arithmetic

Online Class

- [Class 2 - 9 Feb 2026 - Formats, IEEE, Rounding Error](#)
- [Class 3 - 16 Feb 2026 - Rounding Error, Machine Epsilon](#)

Lecture Notes of Class 2 and Class 3 -  Lecture Note Class 2 and 3.pdf 12.8 MiB

Video Lecture

- Class 1  Saadat Rafid Ahmed [MakeupClass - 2025/06/24 - Chapter 01 FloatingPointArithmeti...](#)
- Class 2  Saadat Rafid Ahmed [MakeupClass 2025/06/15 Chapter 01 FloatingPointArithmeti...](#)

Main Topic Lecture Notes (OLD)

-  Floating Point 1.pdf 2869.9KB
-  Floating Point 2.pdf 2530.3KB

While studying consider Denormalised and Normalised to exchange places.

Practice Problems - https://docs.google.com/document/d/1XJgb5TT_Bn2pUTsmAyDTxIwbWgBuMu28hM-Mc8hJx1c/edit?tab=t.0

▼ Additional Resources

- Floating Point Representation -  RAFEED RAHMAN [Floating Point Representation](#)
- IEEE -  Abu Mohammad Khan [Floating-3](#)
- Error Analysis -  RAFEED RAHMAN [Rounding Error](#)

2 Chapter 2: Polynomial Interpolation

Chapter 2 Pre-requisite

- Abu Mohammad Khan Polynomial-1
- Abu Mohammad Khan Polynomial-2
- Abu Mohammad Khan Polynomial-3
- Abu Mohammad Khan Polynomial-4

Online Class

CSE330 - Spring 2026 - Class 6[Makeup] - 27 Feb 2026: Zoom Video Conferencing, Web Conferencing, Webinars, Screen Shari...

CSE330[13] - Spring 2026 - Class 8 - 9 Mar 2026 - Zoom Video Conferencing, Web Conferencing, Webinars, Screen Shari...

Lecture Notes

- Lagrange Polynomial [Lecture Note].pdf 170.6 KiB
- Newton Divided Difference [Lecture Note].pdf 185.1 KiB
- Hermite Polynomial [Lecture Note].pdf 196.7 KiB

- 1 CHAPTER I Polynomial Interpolation.pdf 3249.5KB
 - 2 CHAPTER I Polynomial Interpolation.pdf 916.6KB
 - 3&4 Chapter 2 Newton and Hermite.pdf 5051.2KB
- Polynomial 1.pdf 8600.0KB
 - Polynomial 2.pdf 2404.7KB
 - Polynomial 3.pdf 2621.2KB

Video Lecture

- [Online Class] VM Hermite Cauchy's

- ▼ Additional Practice Problem (OLD)
- ▼ Taylor Series

Question # 1: Consider the function $f(x)=x e^x$. In the following, the interpolating polynomial, $p_3(x)$, is computed by using Taylor expansion. To do so, do the following tasks

- [2 Marks] Using Taylor expansion of $x e^x$, write $f(x)$ as an infinite series.
- [2 Marks] Find the values of $a_0, a_1, \cdots, a_3$ if the function is interpolated by degree three polynomial $p_3(x)$.
- [2 Marks] Compute: $f(0.1)$ and $p_3(0.1)$ up to seven significant figures.
- [2 Marks] Find the percent error for interpolating $f(x)$ by $p_3(x)$.

- ▼ VM & Lagrange Polynomial (Use both the methods)

Sl.	x	Y
1	-1	2.2
2	0	10.6
3	1	17.0
4	2	22.4

a. Use the nodes given above to set up a lagrange polynomial of appropriate degree. Find an approximate value of Pn(2.5)

b. Use the nodes given above to set up a lagrange polynomial of appropriate degree. Find an approximate value of Pn(2.5)

- ▼ Chapter 2 Practice Problem
- Chapter 2 Practice Questions.pdf 369.6KB

▼ NOTE ON CAUCHY’S THEOREM WITH NO INTERVAL

CAUCHY’S THEOREM WITH NO INTERVAL

Suppose you are finding the max truncation error with Cauchy’s Thorem and you want to maximise the value of $\frac{f^{(n+1)}(\xi)}{n!}$, however, you come up accross a term

$$f^{(n+1)}(\xi) = 2cos(\xi) + sin(\xi) \dots (1)$$

So how would you maximise it?

- If the interval [a,b] is provided then you go thorough the regular process of checking the value of the $f^{(n+1)}(\xi)$ at each interval. So, basically the maximim value is $max(f^{(n+1)}(a), f^{(n+1)}(b))$
- If not interval provided, you maximise each component separately, as no value can be larger than that the sum of each of the terms maximum value. So, In the equation(1) we have two terms. $2cos(\xi)$ & $sin(\xi)$. So, the maximum value of $max(2cos(\xi)) = 2 \times cos(0) = 2 \times 1 = 2$. And, $max(sin(\pi/2)) = 1$. So the maximum of

$$max(f^{(n+1)}(\xi)) = 2 \times 1 + 1 = 3$$

3 Chapter 3: Numerical Differentiation

ONLINE CLASS

CSE330[13] - Spring 2026 - Class 9 - 11 Mar 2026 -  Zoom [Video Conferencing, Web Conferencing, Webinars, Screen Shari...](#)

 Richardson Extrapolation Proof General form.pdf 150.4 KiB

Slides - [Chapter3 PPTX](#)

 Chapter 3.pdf 5.1 MB

Video Lecture

 [\[Online Class\] Differentiation](#)

Additional Notes

 1 Chapter 3.pdf 2090.9KB

 Differentiation 1.pdf 2605.4KB

 2 Chapter 3.pdf 1654.6KB

 Differentiation 2.pdf 2579.6KB

▼ Additional Resources

Basics -  Abu Mohammad Khan [Differentiation-1](#)

Forward Differentiation & Backward Differentiation -  Abu Mohammad Khan [Differentiation-2](#)

Central Differentiation -  Abu Mohammad Khan [Differentiation-3](#)

Rounding Error -  Abu Mohammad Khan [Differentiation-4](#)

Richardson Extrapolation -  Abu Mohammad Khan [Differentiation-5](#)

Error Upper Bound -  RAFEED RAHMAN [Upper Bound of Truncation Error in Differenciation](#)

▼ Additional Practice Problem (OLD)

▼ Question 01

Question 1

Question # 1: A function is given by $f(x) = 6e^{-3x}$. Now Answer the following:

- [1 Mark] Calculate $f'(x)$ at $x = 0.5$ with $h = 0.32$ using the central difference formula.
- [1 Mark] Calculate $f'(x)$ at $x = 0.5$ with $h = 0.16$ using the central difference formula.
- [3 Marks] Now compute $D_{0.32}^{(1)}$ at $x = 0.5$ using Richardson extrapolation method.
- [2 Marks] If the exact value of the derivative, $f'(0.5)$ is 1.3388, find the percentage error with extrapolated value found in the previous part.

▼ Question 02

1. Consider the following data set:

x	1.1	1.2	1.3
$f(x)$	-0.57941	-0.90730	-1.2807

(a) (4 marks) Using the above data, compute $f'(1.2)$ using the central difference method.

(b) (6 marks) For in interval $[1.1, 1.3]$, compute the error bound (truncation error) if the above data is generated by the function, $f(x) = x \cos(x) - x^2 \sin(x)$.

(c) (2 marks) Also compute the actual error.

Solution: Here: $h = 0.1$, $x_0 = 1.2$. Therefore, the derivative at $x = 1.2$ is

$$f'(1.2) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} \Big|_{x_0=1.2, h=0.1} = \frac{-1.2807 - (-0.57941)}{2 \times 0.1} = -3.5065 \text{ .}$$

Solution: For $\xi \in [1.1, 1.3]$, the error bound is given by

$$\begin{aligned} \left| \frac{h^2}{6} f'''(\xi) \right| &= \frac{h^2}{6} \left| -9 \cos(\xi) + 7\xi \sin \xi + \xi^2 \cos \xi \right| \text{ ,} \\ &\leq \frac{(0.1)^2}{6} \left(9|\cos(1.1)| + 7(1.3)|\sin(1.3)| + (1.3)^2|\cos(1.1)| \right) \text{ ,} \\ &= 0.022696 \text{ .} \end{aligned}$$
$$f'(1.2) = \left(\cos x - 3x \sin x - x^2 \cos x \right) \Big|_{x=1.2} = -3.5148 \text{ .}$$

$\therefore$ Actual Error = $|-3.5148 - (-3.5065)| = 0.0083 \text{ .}$

